

A Discriminant Lower Bound for Divisibility-Covering Difference Sets, with an Application to the Covering-Pair Exponents

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Abstract

Let S be a finite set of integers with all elements of absolute value at most M . We say S *divisibility-covers* a set I of positive integers if every $i \in I$ divides some nonzero difference $s - s'$ with $s, s' \in S$. We prove, by an elementary discriminant argument, that covering the primes in a dyadic interval $(n/2, n]$ forces $|S|^2 \log M \gtrsim n/2$. In the exponent scaling $|S| = n^\beta$, $M = \exp(n^\alpha)$, this reads $2\beta + \alpha \geq 1$. The proof identifies the covering condition with divisibility of the Vandermonde discriminant of the polynomial $\prod_{s \in S} (x - s)$ by the primorial of the interval, and the bound is the magnitude comparison $|\text{disc}| \geq \text{primorial}$. We include a remark delimiting what the method does and does not give: it is essentially tight as a *magnitude* bound and cannot by itself yield any stronger exponent inequality.

1 Statement

For a finite set $S \subset \mathbb{Z}$ write $\Delta(S) = \{|s - s'| : s, s' \in S, s \neq s'\}$ for its set of positive differences.

Definition 1. Let $I \subset \mathbb{Z}_{>0}$. We say S *divisibility-covers* I if for every $i \in I$ there exists $d \in \Delta(S)$ with $i \mid d$.

Throughout, $n \geq 4$ and $M \geq 2$ are real parameters, and

$$P(n) = \{p \text{ prime} : n/2 < p \leq n\}, \quad Q(n) = \prod_{p \in P(n)} p.$$

Theorem 1. Let $S \subset \mathbb{Z}$ be finite with $|s| \leq M$ for all $s \in S$, and suppose S *divisibility-covers* $P(n)$. Then

$$|S| (|S| - 1) \log(2M) \geq \log Q(n) = \theta(n) - \theta(n/2),$$

where θ is the first Chebyshev function. In particular, since $\theta(n) - \theta(n/2) = \frac{n}{2} + o(n)$,

$$|S|^2 \geq \frac{n}{2 \log(2M)} (1 + o(1)).$$

Corollary 1. Suppose a family of covering sets satisfies $|S| = n^{\beta+o(1)}$ and $M = \exp(n^{\alpha+o(1)})$ with $0 \leq \alpha < 1$. Then

$$2\beta + \alpha \geq 1.$$

2 Proof

Let $S = \{s_1, \dots, s_K\}$ with $K = |S|$ and form the monic polynomial

$$f(x) = \prod_{k=1}^K (x - s_k) \in \mathbb{Z}[x].$$

By the product-and-sum rule, $f'(x) = \sum_k \prod_{j \neq k} (x - s_j)$, and the discriminant of f satisfies the classical identities

$$\text{disc}(f) = (-1)^{K(K-1)/2} \text{Res}(f, f') = \prod_{i < j} (s_i - s_j)^2. \quad (1)$$

Lemma 1. *For a prime p , we have $p \mid \text{disc}(f)$ if and only if some nonzero difference $s_i - s_j$ is divisible by p ; equivalently, iff f has a repeated root modulo p .*

Proof. By (1), $p \mid \text{disc}(f)$ iff $p \mid \prod_{i < j} (s_i - s_j)^2$, iff $p \mid s_i - s_j$ for some $i < j$ (as p is prime). The same condition says two roots of f coincide modulo p , i.e. f has a repeated root mod p . $\square$

Now suppose S divisibility-covers $P(n)$. Fix $p \in P(n)$. Since $p > n/2 \geq 2$ and p divides some $d = |s_i - s_j| \in \Delta(S)$, Lemma 1 gives $p \mid \text{disc}(f)$. As the primes in $P(n)$ are distinct, their product divides $\text{disc}(f)$:

$$Q(n) = \prod_{p \in P(n)} p \mid \text{disc}(f). \quad (2)$$

We now compare magnitudes. First, $\text{disc}(f) \neq 0$: the s_k are distinct integers, so every factor $s_i - s_j$ in (1) is nonzero, hence $\text{disc}(f)$ is a nonzero integer. From (2) and $\text{disc}(f) \neq 0$,

$$|\text{disc}(f)| \geq Q(n). \quad (3)$$

Second, each $|s_i - s_j| \leq |s_i| + |s_j| \leq 2M$, so by (1),

$$\log |\text{disc}(f)| = 2 \sum_{i < j} \log |s_i - s_j| \leq 2 \binom{K}{2} \log(2M) = K(K-1) \log(2M). \quad (4)$$

Combining (3) and (4),

$$K(K-1) \log(2M) \geq \log Q(n) = \sum_{p \in P(n)} \log p = \theta(n) - \theta(n/2),$$

which is the first display of the theorem. By the prime number theorem in the form $\theta(x) = x + o(x)$ we have $\theta(n) - \theta(n/2) = \frac{n}{2} + o(n)$, giving $K^2 \geq \frac{n}{2 \log(2M)} (1 + o(1))$.

For Corollary 1, substitute $K = n^{\beta+o(1)}$ and $\log(2M) = n^{\alpha+o(1)}$: then $n^{2\beta+o(1)} \geq n^{1-\alpha+o(1)}$, i.e. $2\beta + \alpha \geq 1$. $\square$

3 Scope of the method

Remark 1 (The bound is tight *as a magnitude bound*). Inequality (4) is sharp to leading order: for K points spread in $[0, M]$, the maximum of $\sum_{i < j} \log |s_i - s_j|$ is $\frac{K^2}{2} \log M - O(K^2)$ (attained near Fekete/Chebyshev configurations). Hence no choice of the positions of the s_k improves the exponent in Theorem 1: the additive structure of a difference set affects only the $O(K^2)$ lower-order term, not the leading $\frac{K^2}{2} \log M$. Consequently the discriminant magnitude comparison *cannot* yield any exponent bound stronger than $2\beta + \alpha \geq 1$.

Remark 2 (What is *not* proved). Theorem 1 is consistent with, but does not establish, the stronger inequality $\alpha + \beta \geq 1$ (which would require $\beta \geq 1 - \alpha$, roughly twice the exponent proved here). Whether $2\beta + \alpha \geq 1$ is attained – equivalently, whether divisibility-covering sets exist with $|S| = n^{(1-\alpha)/2+o(1)}$ – is left open. Establishing either a matching construction or a strictly stronger lower bound appears to need input beyond the magnitude of the discriminant; heuristic considerations (a sum–product antagonism between additive economy and the multiplicative richness of a difference set) suggest the honest frontier lies there.

Remark 3 (Covering the full interval). If S divisibility-covers all of $(n/2, n] \cap \mathbb{Z}$ (not merely the primes), the hypothesis of Theorem 1 holds a fortiori, so the same lower bound applies.